

Ashok Vardhan Makuva

CONTACT	Email: ashokevardhan@gmail.com	Homepage: https://ashokvardhan.github.io/
EDUCATION	University of Illinois at Urbana-Champaign (UIUC) Ph.D., Electrical and Computer Engineering, 2017 - 2022 4.0/4.0 – Advisor: Pramod Viswanath M.S., Electrical and Computer Engineering, 2015 - 2017 4.0/4.0 – Advisor: Yihong Wu Indian Institute of Technology Bombay (IIT Bombay) B.Tech., Electrical Engineering, 2011 - 2015 9.62/10.0 – Advisor: Vivek Borkar	
PROFESSIONAL APPOINTMENTS	• Associate Professor – Télécom Paris, IP Paris [Nov 25 –] <i>Department of Computer Science, Data and AI</i> • Postdoctoral Researcher – EPFL [Sep 22 – Sep 25] <i>Department of Computer and Communication Sciences</i> Mentor: Michael Gastpar	
GRANTS	• EuroTech Visiting Researcher, with myself as the sole PI [Jul - Aug 2026] Funded research visits to top European universities to foster new collaborations	
LEADERSHIP	Co-organizer and Presenter – NeurIPS 2024 Tutorial, Sandbox for the Blackbox • Led the design, coordination, and presentation of cutting-edge content on the novel structured sandbox approach to demystify black-box LLMs	
SELECT AWARDS	• ICLR Oral: From Markov to Laplace (1% out of 18,949 papers) [2026] • DAAD Ainet Fellowship: Awarded to top AI researchers to visit German universities [2025] • NeurIPS Spotlight: What One Cannot, Two Can (3% out of 21,575 papers) [2025] • ICLR Spotlight: Attention with Markov (5% out of 11,670 papers) [2025] • Best Paper Award: ACM Mobihoc [2019] • Joan and Lalit Bahl Fellowship, UIUC (awarded twice) [2019, 2020] • Sundaram Seshu International Student Fellowship, UIUC [2018] • Qualcomm Innovation Fellowship Finalist (among 174 applicants) [2018] • All India Rank 32: Awarded fellowship in IISc for undergraduate studies (declined) [2011]	
TEACHING	COM105 “Digital Communications and Information Theory”, Télécom Paris [2026] Graduate Teaching Assistant : 3 semesters at UIUC, 5 semesters at IIT Bombay [2013 – 2020] • UIUC: Information Theory (ECE 563), Representation Learning (ECE 598), Detection and Estimation Theory (ECE 561) • IIT Bombay: Linear Algebra (MA 106), Differential Equations I-II (MA 108, MA 208), Complex Analysis (MA 205) & Electricity and Magnetism (PH 103)	
SELECT TALKS	1. From Markov to Laplace: A Markovian Tale of LLMs [2025 – 2026] • ENS Paris & Inria Paris, <i>Host</i>: Francis Bach • ENS Ulm, <i>Host</i>: Gabriel Peyré • Stanford University, <i>IT Forum</i> • ETH Zürich, <i>Hosts</i>: Andreas Krause, Thomas Hofmann • École Polytechnique, <i>Applied Mathematics Seminar</i> • FLaNN Seminar & ITA Workshop, <i>San Diego</i>	

2. **Sandbox for the Blackbox: How LLMs Learn Structured Data** [2025]
 - ITCS tutorial, *Data Science Seminar*
3. **KO codes** [2021 – 2022]
 - Talks at top institutions worldwide, including MIT, Stanford, Berkeley, CMU, UCSD, UToronto, ETH Zürich, EPFL, IST Austria, TIFR, and IISc.
4. **Learning in Gated Neural Networks** [2018 – 2020]
 - Talks at leading institutions in the US and India, including CMU, MSR, IIT Madras, IIT Bombay, and TIFR.

SELECT
PUBLICATIONS

1. **From Markov to Laplace: How Mamba In-Context Learns Markov Chains**
M. Bondaschi, N. Rajaraman, X. Wei, R. Pascanu, C. Gulcehre, M. Gastpar, **A.V. Makkuva**
ICLR, 2026 (Oral, 1% out of 18,949 papers).
2. **What One Cannot, Two Can: Two-Layer Transformers Provably Represent Induction Heads on Any-Order Markov Chains**
C. Ekbote, M. Bondaschi, N. Rajaraman, J. D. Lee, M. Gastpar, P. P. Liang*, **A.V. Makkuva***
NeurIPS, 2025 (Spotlight, 3% out of 12,575 papers).
3. **Fundamental Limits of Prompt Compression: A Rate-Distortion Framework for Black-Box Language Models**
A. Girish, A. Nagle, M. Bondaschi, M. Gastpar, H. Kim*, **A.V. Makkuva***
NeurIPS, 2024.
4. **Optimal transport mapping via input convex neural networks**
A.V. Makkuva*, A. Taghvaei*, J.D. Lee, S. Oh
ICML, 2020.
5. **Barracuda: The Power of ℓ -polling in Proof-of-Stake Blockchains**
G. Fanti, J. Jiao, **A.V. Makkuva**, S. Oh, R. Rana, P. Viswanath
ACM Mobihoc, 2019 (Best paper award).

FULL
PUBLICATION
LIST

21. **TERMINATOR: Learning Optimal Exit Points for Early Stopping in CoT Reasoning**
A. Nagle, J. Saydaliev, D. Garbaya, M. Gastpar, H. Kim*, **A.V. Makkuva***
TTU Workshop, ICLR 2026.
20. **From Markov to Laplace: How Mamba In-Context Learns Markov Chains**
M. Bondaschi, N. Rajaraman, X. Wei, R. Pascanu, C. Gulcehre, M. Gastpar, **A.V. Makkuva**
ICLR, 2026 (Oral, 1% out of 18,949 papers).
19. **What One Cannot, Two Can: Two-Layer Transformers Provably Represent Induction Heads on Any-Order Markov Chains**
C. Ekbote, M. Bondaschi, N. Rajaraman, J. D. Lee, M. Gastpar, P. P. Liang*, **A.V. Makkuva***
NeurIPS, 2025 (Spotlight, 3% out of 12,575 papers).
18. **Attention with Markov: A Curious Case of Single-layer Transformers**
A.V. Makkuva*, M. Bondaschi*, A. Girish, A. Nagle, M. Jaggi, H. Kim, M. Gastpar
ICLR, 2025 (Spotlight, 5% out of 11,670 papers).
17. **Fundamental Limits of Prompt Compression: A Rate-Distortion Framework for Black-Box Language Models**
A. Girish, A. Nagle, M. Bondaschi, M. Gastpar, H. Kim*, **A.V. Makkuva***
NeurIPS, 2024.
16. **Transformers on Markov Data: Constant Depth Suffices**
N. Rajaraman, M. Bondaschi, K. Ramchandran, M. Gastpar, **A.V. Makkuva**
NeurIPS, 2024.
15. **Local to Global: Learning Dynamics and Effect of Initialization for Transformers**
A.V. Makkuva*, M. Bondaschi*, C. Ekbote, A. Girish, A. Nagle, H. Kim, M. Gastpar
NeurIPS, 2024.
14. **LASER: Linear Compression in Wireless Distributed Optimization**
A.V. Makkuva*, M. Bondaschi*, T. Vogels, M. Jaggi, H. Kim, M. Gastpar
ICML, 2024.

13. **CRISP: Curriculum based Sequential Neural Decoders for Polar Code Family**
S.A. Hebbar*, V. Nadkarni*, **A.V. Makkuva**, S. Bhat, S. Oh, P. Viswanath
ICML, 2023.
12. **Machine Learning-Aided Efficient Decoding of Reed-Muller Subcodes**
M.V. Jamali, X. Liu, **A.V. Makkuva**, H. MahdaviFar, S. Oh, P. Viswanath
IEEE Journal on Selected Areas in Information Theory (JSait), 2023.
11. **TinyTurbo: Efficient Turbo Decoders on Edge**
S.A. Hebbar*, R. Mishra*, S.K. Ankireddy, **A.V. Makkuva**, H. Kim, P. Viswanath
IEEE International Symposium on Information Theory (ISIT), 2022.
10. **KO codes: Inventing Nonlinear Encoding and Decoding for Reliable Wireless Communication via Deep-learning**
A.V. Makkuva*, X. Liu*, M.V. Jamali, H. MahdaviFar, S. Oh, P. Viswanath
ICML, 2021.
9. **Reed-Muller Subcodes: Machine Learning-Aided Design of Efficient Soft Recursive Decoding**
M.V. Jamali, X. Liu, **A.V. Makkuva**, H. MahdaviFar, S. Oh, P. Viswanath
ISIT, 2021.
8. **Optimal transport mapping via input convex neural networks**
A.V. Makkuva*, A. Taghvaei*, J.D. Lee, S. Oh
ICML, 2020.
7. **Learning in Gated Neural Networks**
A.V. Makkuva, S. Oh, S. Kannan, P. Viswanath
AISTATS, 2020.
6. **Breaking the gridlock in Mixture-of-Experts: Consistent and Efficient Algorithms**
A.V. Makkuva, S. Oh, S. Kannan, P. Viswanath
ICML, 2019.
5. **Barracuda: The Power of ℓ -polling in Proof-of-Stake Blockchains**
G. Fanti, J. Jiao, **A.V. Makkuva**, S. Oh, R. Rana, P. Viswanath
ACM International Symposium on Mobile Ad Hoc Networking and Computing (ACM Mobihoc), 2019 (Best paper award).
4. **Learning One-hidden-layer Neural Networks under General Input Distributions**
W. Gao*, **A.V. Makkuva***, S. Oh, P. Viswanath
AISTATS, 2019.
3. **Equivalence of additive-combinatorial linear inequalities for Shannon entropy and differential entropy**
A.V. Makkuva, Y. Wu
IEEE Transactions on Information Theory, 2018.
2. **On additive-combinatorial affine inequalities for Shannon entropy and differential entropy**
A.V. Makkuva, Y. Wu
ISIT, 2016.
1. **Event-driven stochastic approximation**
N. Sahasrabudhe, **A.V. Makkuva**, V.S. Borkar
Indian Journal of Pure and Applied Mathematics, 2016.

STUDENTS

PhD advisees

- [Rami Mohammadamin](#) (co-advised with Aslan Tchamkerten)
– Expected Graduation (8/15/2029)
- [Amirmehdi Jafari Fesharaki](#) (co-advised with Aslan Tchamkerten)
– Expected Graduation (8/15/2028)

Mentees

- [Chanakya Ekbote](#) (PhD at MIT)
Publications (19, 15)

- [Alliot Nagle](#) (PhD at UT Austin)
Publications (21, 18, 17, 15)
- [Jakhongir Saydaliev](#) (MS at EPFL & UT Austin)
Publications (21)
- [Dhia Garbaya](#) (MS at École des Ponts & ENS Paris-Saclay)
Publications (21)
- [Marco Bondaschi](#) (PhD at EPFL)
Publications (20, 19, 18, 17, 16, 15, 14)
- [Nived Rajaraman](#) (PhD at UC Berkeley → Postdoc at MSR NYC)
Publications (20, 19, 16)
- [Adway Girish](#) (PhD at EPFL)
Publications (18, 17, 15)
- [Thijs Vogels](#) (PhD at EPFL → MSR Amsterdam)
Publications (14)
- [Ranvir Rana](#) (PhD at UIUC → Co-founder & CTO at Kaleidoscope Blockchain)
Publications (5, **Best paper award**)
- [Xiyang Liu](#) (PhD at University of Washington)
Publications (12, 10, 9) **Qualcomm Fellowship Winner**
- [Mohammad Vahid Jamali](#) (PhD at U. Michigan → Samsung)
Publications (12, 10, 9) **Qualcomm Fellowship Winner**
- [Ashwin Hebbar](#) (MS at UIUC → PhD at Princeton)
Publications (13, 11)
- [Viraj Nadkarni](#) (MS at UIUC → PhD at Princeton)
Publications (13)
- [Sravan Kumar Ankireddy](#) (PhD at UT Austin)
Publications (11)

ACADEMIC SERVICE

Reviewer

[2015 –]

- Conferences: NeurIPS, ICML, ICLR, AISTATS, ISIT

SCHOLASTIC ACHIEVEMENTS

- Secured **10/10** GPA at IIT Bombay, Spring 2014 - 2015
- Secured **All India Rank 14** in 41st National Mathematical Talent Competition
- Secured **All India Rank 32** in AIEEE among 10,65,100 students
- Secured **All India Rank 287** in IIT-JEE among 4,85,000 students

[2010]

[2011]

[2011]

PATENTS

- **Non-linear encoding and decoding for reliable wireless communication**
A.V. Makkuva, X. Liu, M.V. Jamali, H. MahdaviFar, S. Oh, P. Viswanath

[2022]

[[google patents](#)]